THE DELTA INVARIANT BOUNDS THE NUMBER OF GEOMETRIC BRANCHES MINUS ONE

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- Delta invariant unchanged under strict henselization. delivered companion lemma-delta-same-after-completion. First input of the reduction.
- Delta invariant versus geometric branches in the strictly henselian case. delivered companion lemma-delta-number-branches-inequality-sh. The case actually invoked.
- Number of geometric branches. More on Algebra, Definition more-algebra-definition-number-of-branches. Definitional invariance under strict henselization.

Source reference. Original lemma 0C3W: sources/stacks-chow/tags.txt, source lines 15206–15206. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0C26: sources/stacks-chow/tags.txt, source lines 15146–15146. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C42: sources/stacks-chow/tags.txt, source lines 15212–15212. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a 1-dimensional reduced Nagata local ring. Then*

$$\delta\text{-invariant of } A \geq \text{number of geometric branches of } A - 1$$

Proof. We may replace A by the strict henselization of A without changing the δ-invariant (Lemma 0C3W) and without changing the number of geometric branches of A (this is immediate from the definition, see More on Algebra, Definition 0C26). Thus we may assume A is strictly henselian and we may apply Lemma 0C42. □

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